Aman Sharma

Machine Learning Engineer/ AI Engineer/ Datascientist/ Research Engineer

linkedin.com/in/aman-sharma-8688a81b0
github.com/dr-aman-sharma aman9716323072@gmail.com
New Delhi, India
https://orcid.org/0009-0005-9493-0632
Google Scholar link

I am transitioning into ML/AI engineering, with a focus on building practical, production-relevant AI systems. I have developed/developing a portfolio of projects spanning computer vision, natural language processing, agentic AI systems, RAG and fine-tuning. My background as a computation expert and analyst allows me to approach AI problems with a strong focus on modeling, rigor, and system-level thinking. It also provides me an ability to contribute to building robust, real-world AI applications.

Skills

Languages : Python (expert), Julia, MATLAB, R; **Machine Learning :** PyTorch, TensorFlow, Scikit-learn, Hugging Face; Deep Learning : CNN, Transformers, NLP (ROBERTA), Computer Vision (UNET, YOLOX), Generative AI, Fine-tuning (QLoRA, LoRA, Adapters); **Data Analysis :** Pandas, NumPy, SciPy, SQL, Excel; **Visualization :** Matplotlib, Seaborn, Tableau; Parallel Computing : OpenMP, MPI, HPC (Slurm); **Cloud :** AWS, IBM Watson Studio; **Tools :** Git, OpenCV; **Additional :** Scientific writing, project supervision, teaching

Top AI Projects (https://github.com/dr-aman-sharma)

- **Comment Sentiment Classification | ML Model Comparison & Hyperparameter Tuning :**
  Trained and compared multiple machine learning models to classify user comments into positive, negative, and neutral categories. Conducted systematic hyperparameter tuning across diverse algorithms to identify the optimal configuration. Retrained the best-performing model on the full dataset for final evaluation.

- **Pixel-Wise Image Segmentation | Generalizable Clothing Classification :**
  Developed a semantic segmentation model assigning each pixel to a specific clothing category. Built on a clothing segmentation dataset with ground-truth masks. Designed the framework to be easily adaptable to other image segmentation datasets (e.g., Cityscapes, medical imaging) with minimal modifications.

- **Discourse Classification with NER | Rhetorical Role Labeling in Text**
  Built a token-level sequence labeling model (NER-style) to classify text segments into seven discourse categories : Lead, Position, Claim, Counterclaim, Rebuttal, Evidence, and Concluding Statement. Trained on thousands of fully annotated argumentative texts where a single document contains multiple overlapping rhetorical functions.

- **YOLO Object Detector**
  Built end-to-end object detection system from scratch using TensorFlow/Keras with custom loss function, EfficientNetB1 backbone, and NMS inference pipeline (PASCAL VOC 2012, 20 classes, 17k+ images).

- **AI Coding Agent | Gemini Function Calling for Autonomous Debugging**
  Built an autonomous coding agent using Google's Gemini API that reads directories, reads and writes files, and executes programs. The agent uses function calling to interact with a codebase, analyze outputs, and autonomously fix bugs through a complete feedback loop of observation, action, and verification.

Professional Certificates

- **IBM AI Engineering Professional Certificate** (March 2026) — 13 courses including Deep Learning, PyTorch, TensorFlow, LangChain, RAG, and LLMs (https://coursera.org/share/a2b76434c4244a7aaceaaa50b910e413)

- **IBM Data Science Professional Certificate** (April 2026) — 12 courses including Python, SQL, Data Analysis, Visualization, and Machine Learning (https://coursera.org/share/1fc4605d1887e5bec749eea20f997c2f)

Professional Experience

Present – November 2025

Machine Learning Engineer (Projects), Self-directed/ Independent, New Delhi, India

- Developed ML systems across NLP, CV, and LLM applications.
- Built transformer-based NLP models and CV pipelines (segmentation, detection).
- Implemented RAG system and LLM-based coding agent.
- Conducted fine-tuning (LoRA/QLoRA) and model evaluation.

February 2026 – September 2021

Computation Expert and analyst, École Polytechnique Fédérale de Lausanne (EPFL), Lausanne, Switzerland

- Developed custom **TEBD** and **DMRG** codes to simulate spin dynamics in frustrated Heisenberg chains.
- Studied confinement, deconfinement, and bound states in various spin models using MPS-based simulations.
- Analyzed inelastic neutron scattering data on QSL candidate $K_2Ni_2(SO_4)_3$ using **Horace** software.
- Published 3 peer-reviewed articles in *Phys. Rev. B*, and 1 in *Nature Communications*.

2017 – 2021

Research Internships & Master's Thesis – Computational Condensed Matter, IISER Mohali & IISc Bangalore, India

- Simulated topological quantum systems, including Majorana modes and phase transitions in nanowires using **KWANT**.
- Developed numerical simulations for strongly correlated systems (e.g., Wigner crystals) using cluster mean-field methods.
- Studied superconducting qubits and designed theoretical quantum circuits based on Josephson junctions.
- Built mathematical foundations in topology and metric spaces, and conducted experimental work on light polarization.

Education

2016 – 2021 Bachelor of Science and Master of Science (BS-MS Dual Degree) in Physics, Indian Institute of Science Education and Research (IISER) Mohali, India. CPI : 9.1/10. Thesis : Majorana Fermions in Superconductor–Semiconductor Heterostructures.

2015 Senior Secondary (CBSE), Delhi, India. Score : 93% – Physics, Mathematics, Computer Science, Chemistry, English.

IIT-JEE Mains & Advanced qualified (2016), IAPT-NGPE exam all India top 1 percent (2018), CSIR-NET exam qualified - all India, 99.2 percentile (2019)

Publications (https://orcid.org/0009-0005-9493-0632)

- M. G. Gonzalez, V. Noculak, **A. Sharma**, et al., *Dynamics of $K_2Ni_2(SO_4)_3$ governed by proximity to a 3D spin liquid model*, *Nature Communications*, **15**, 7191 (2024).
- **A. Sharma**, M. Nayak, H. M. Rønnow, F. Mila, *Bound states and deconfined spinons in the dynamical structure factor of the $J_1 - J_2$ spin-1 chain*, *Phys. Rev. B*, **111**, 064404 (February, 2025).
- **A. Sharma**, M. Nayak, N. Chepiga, F. Mila, 2025. *Excitations and dynamical structure factor of $J_1 - J_2$ spin-3/2 and spin-5/2 Heisenberg spin chains*, *Phys. Rev. B*, **112**, 104401 (September, 2025).
- **A. Sharma**, M. Nayak, N. Chepiga, H. M. Rønnow, F. Mila, *Confinement, deconfinement, and bound states in the spin-1 and spin-$3/2$ generalizations of the Majumdar–Ghosh chain*, *Phys. Rev. B*, **113**, 054438 (February, 2026).

Languages

English (Fluent) ●●●●●

Hindi (Native) ●●●●●

French ●●○○○

Sanskrit ●●○○○

Strengths

- Expertise in scientific computing and simulation
- Strong analytical and problem-solving mindset
- Ability to communicate complex concepts clearly
- Self-driven and research-focused